5.18. THE HEISENBERG UNCERTAINTY PRINCIPLE

In this chapter we have repeatedly stated that the nuclear decay energies are exact values as required by quantum mechanics. However, this is not exactly correct: the energy levels have a certain "spread". This was first stated by Heisenberg in 1927 and is of fundamental importance in all areas of nuclear physics.

The *uncertainty principle* states that it is impossible to measure simultaneously the exact position and the exact momentum of a particle. This follows from the wave properties of the particle. If for example, we attempt to measure the exact position of an electron by observing the light emitted when it hits a scintillating screen, this act interferes with the movement of the electron causing it to scatter, which introduces some uncertainty in its momentum. The size of this uncertainty can be calculated exactly and is related to the Planck constant. If Δx denotes the uncertainty in position and Δp the uncertainty in momentum along the x axis, then

$$\Delta x \, \Delta p \geq \mathbf{h}/(2\pi) \equiv \hbar \tag{5.87}$$

$\hbar$ is called "h-bar" (1.05×10^{-34} J s), and $\mathbf{h}$ is the *Planck constant*.

This principle holds for other *conjugate variables*, as angle θ and angular momentum p_θ

$$\Delta\theta \, \Delta p_\theta \geq \hbar \tag{5.88}$$

and time and energy

$$\tau \, \Delta E \geq \hbar \tag{5.89}$$

This latter equation relates the life-time τ of an elementary (or nuclear) particle to the uncertainty in its energy (ΔE). For excited nuclear states this can be taken as the width of the γ-peak at half-maximum intensity (the "FWHM value") (Fig. 5.14). For example, if $\Delta E_\gamma = 1.6$ keV, then $\tau \geq 1.05 \times 10^{-34}/1600 \times 1.60 \times 10^{-19}$ s $= 4.1 \times 10^{-19}$ s. This is a

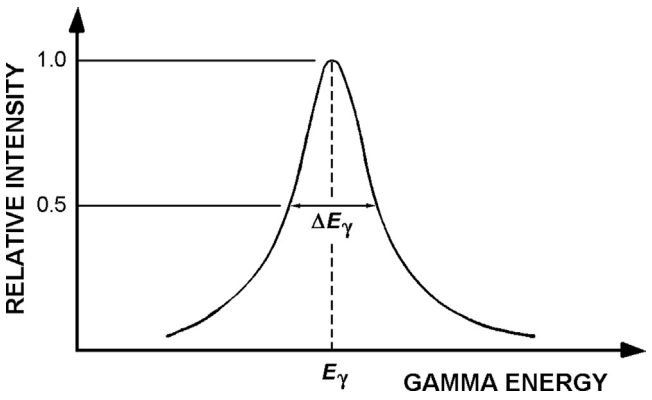


Figure 5.14 The energy half-width value ΔE_γ is the FWHM of the γ-peak (FWHM = Full Width at Half Maximum).

long time compared to that of a nuclear rotation, which is about 10^{-23} s. Consequently the wave mechanic properties of particles (and γ-rays) introduce a certain fundamental uncertainty in the particle energy. Several applications of this are described in later chapters.

5.19. EXERCISES

For some of the problems necessary nuclear data are given in the Tables or appendices.

5.1. ^{239}Pu emits α-particles of maximum 5.152 MeV. What is the recoil energy of the product formed?

5.2. Using a magnetic spectrometer the maximum energy of the electrons from ^{137}Cs was found in Figure 5.1 to correspond to 3.15×10^{-3} Tesla m. Calculate the energy (a) assuming that the electrons are non-relativistic, (b) with correction for relativistic mass increase.

5.3. ^{11}C decays through emission of positrons of a maximum energy of 1.0 MeV. Calculate the recoil energy of the daughter.

5.4. ^{16}N decays through β^- decay to ^{16}O with a half-life of 7.1 s. A number of very energetic γ's follow after the β-emission, the dominating one with an energy of 6.14 MeV. What is the ^{16}O recoil energy?

5.5. The binding energy of a K-electron in barium is 37 441 eV. Calculate from Figure 5.1 the internal conversion energy for ^{137m}Ba (Fig. 5.5).

5.6. From the specific activity of potassium (1850 dpm/g K) and the fact that it all originates in the rare isotope ^{40}K (0.0117%), calculate the half-life of ^{40}K.

5.7. One may assume that when ^{238}U was formed at the genesis an equal amount of ^{235}U was formed. Today the amount of ^{238}U is 138 times the amount of ^{235}U. How long a time ago did the genesis occur according to this assumption?

5.8. The interior of the earth is assumed to be built up of a solid core (radius 1371 km) followed by a molten core (radius 3471 km) and a molten mantle (radius 6354 km) covered by a 17 km thick crust. One assumes that 2% by weight of the molten mantle and crust is potassium; the average mantle density is assumed to be 6000 kg m^{-3} and that of the crust 3300 kg m^{-3}. What energy outflow will the radioactive decay of this element cause at the earth's surface? The decay scheme of ^{40}K is given in Eqn. (13.7). For the EC branch $Q = 1.505$ MeV, for the β^- branch 1.312 MeV. Each decay by the EC branch leads to emission of a 1.46 MeV γ. Compare this energy output to the solar energy influx to the earth of 3.2×10^{24} J y^{-1}.

5.9. A hospital has a 1.5 Ci source of ^{226}Ra in the form of a $RaBr_2$ solution. If the ^{222}Rn is pumped out each 48 h, what is (a) the radon activity (in Bq) at that moment, (b) the radon gas volume at STP?

5.10. (a) Prove the correctness of Eqn. (5.25) by using Newton's laws of motion. (b) Prove the correctness of Eqn. (5.48).